

Valentino Cattaneo

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EDUCATION

Purdue University, College of Engineering | GPA: 3.94/4.00

West Lafayette, IN

Bachelor of Science in Electrical Engineering

August 2024 – May 2028

Concentration: Electric Power and Energy Systems

PROFESSIONAL EXPERIENCE

Angel Aerial Systems

Cincinnati, OH

Electrical Engineering Intern

May 2026 – August 2026

- Redesigned the Trio Scout Battery Management System (BMS) schematic and board shape, adding external cell balancing, autonomous storage discharge and onboard flash memory using Altium
- Led BMS-side root cause analysis of a battery failure, fixing an SMBus timing bug in open-source drivers
- Developed embedded software in NuttX to detect, log and report pack faults to the operator and Angel servers
- Designed the spot-welding assembly for the pouch-cell pack and integrated the new BMS board using NX
- Analyzed 6000 flight logs in Python and identified 70 flight with cell imbalances, driving balancing requirements

Purdue Power Electronics and Drives Research Lab

West Lafayette, IN

Undergraduate Research Assistant

June 2025 – December 2025

- Developed a 6-phase Field-Oriented Control (FOC) motor drive prototype for a UAV propulsion system using two STM32G4 motor drive evaluation boards
- Designed and routed a magnetic encoder PCB in Altium, enabling 14-bit rotor position feedback over SPI
- Modeled two synchronized FOC control loops in Simulink, validating control design before hardware bring-up
- Tuned and validated current- and speed-loop PI gains on a propeller-based propulsion test stand

TECHNICAL PROJECTS AND LEADERSHIP

Vertical Flight Systems Purdue, GoAERO VTOL Competition

West Lafayette, IN

President

August 2025 – Present

- Led a 30-member team developing an autonomy-enabled 248 kg eVTOL, winning a NASA Innovation Award
- Programmed an aircraft sizing algorithm in MATLAB that iterated, converged on and evaluated +1000 battery chemistry, motor, propeller, and multicopter configurations, optimizing aircraft payload capability by 8%
- Co-authored a published conference paper at AIAA AVIATION 2026 on eVTOL development and flight testing
- Engineered the demonstrator's flight test campaign, from initial ground tests through fully autonomous flights

Vice President

August 2024 – August 2025

- Designed and routed a STM32-based 14.4 kWh, 100 V BMS with SoC estimation, balancing and fault detection
- Characterized and selected Li-ion cells and sized packs to meet energy, discharge, and mass requirements
- Selected and integrated the avionics stack and the propulsion system for a 32 kg PX4-based prototype
- Routed, soldered and bench-tested all avionics and power circuits of the prototype

Autonomous 3D-printed Indoor Multirotor – Personal Project

May 2025 – August 2025

- Programmed a LiDAR obstacle detection and mapping, weighted A*, and frontier-based exploration algorithm in C++, enabling autonomous GNSS-denied indoor flight on a PX4 avionics stack and a Raspberry Pi

TECHNICAL SKILLS

Design & Simulation: Altium, KiCAD, LTspice, Simulink, NX, Fusion360

Embedded: STM32 toolchain (CubeMX, CubeIDE, CubeProgrammer), RTOS development, SPI, I²C, CAN, Git

Programming: C, C++, Python, MATLAB